

Realisation of Quantum Unitary Gates using Born Scattering

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Abstract—We develop the Born series (Lippmann–Schwinger) formalism for potential scattering and the associated Møller wave operators and scattering matrix $S(E)$, with the eventual aim of using the scattering matrix at fixed energy as a candidate realisation of a unitary quantum gate acting on plane-wave momentum states. The present manuscript works out the underlying scattering theory in full and shows that $S(E)$ is unitary on the space of fixed-energy states $L^2(S^2)$; a proposed route from $S(E)$ to an explicit finite-dimensional gate matrix is outlined in the concluding section, together with the potential-design problem this raises.

I. INTRODUCTION

Quantum gate synthesis is conventionally addressed either by engineering a time-dependent Hamiltonian whose unitary propagator implements a target gate [2], or by decomposing a target unitary into a sequence of elementary gates realised on a fixed physical platform (superconducting qubits, trapped ions, linear optics, etc.). Both routes treat the interaction as something that is switched on for a controlled duration.

Potential scattering suggests a third route. A particle prepared in a definite-momentum (plane-wave) state $|\psi_i\rangle$ and sent through a static potential $V(r)$ emerges, at large times, in a state $|\psi_0\rangle = S(E)|\psi_i\rangle$, where $S(E)$ is the energy-shell scattering matrix. Because $S(E)$ is unitary on the space of fixed-energy states [4], it is natural to ask whether a suitably designed potential $V(r)$ can make $S(E)$, restricted to a finite-dimensional subspace spanned by a discrete set of incident directions $\{\hat{n}_1, \dots, \hat{n}_N\}$, coincide with a target N -dimensional unitary gate. In this picture the “gate” is not switched on and off in time; it is encoded once, permanently, in the shape of $V(r)$, and different logical basis states are represented by different incident directions (or, equivalently, different points on the momentum sphere S^2) at a fixed energy $E = \hbar^2 k^2 / 2m$.

This is closely related in spirit to scattering-based approaches to quantum information processing that encode qubits in scattering channels rather than in internal degrees of freedom [7], and it complements Fourier/phase-estimation-based constructions of standard gate sets [3]. The present paper carries out the necessary scattering-theory groundwork: Section II derives the Born series for ψ_f order by order in the coupling strength ϵ , resums it into the exact resolvent form,

sets up the Møller wave operators $\Omega_{\pm}(E)$, and obtains the scattering matrix $S(E) = \Omega_+(E)^* \Omega_-(E)$ together with its plane-wave matrix elements $\langle \psi_{i2} | S(E) | \psi_{i1} \rangle$, both in the far-field (radiation-zone) limit and without that approximation. Section III discusses how these matrix elements are proposed to be assembled into an explicit gate matrix, and what remains to be shown.

II. PROBLEM FORMULATION

Consider an incident projectile plane-wave state

$$\psi_i(r) = C \exp(jk\hat{n}_i \cdot r), \quad (1)$$

with projectile energy

$$E = \frac{\hbar^2 k^2}{2m}, \quad (2)$$

where $\hat{n}_i$ is the incident direction. This satisfies the free Schrödinger equation

$$-\frac{\hbar^2}{2m} \Delta \psi_i = E \psi_i. \quad (3)$$

A. Born Series

Let the interaction potential energy be $\epsilon V(r)$ and write the final (scattered) state as a formal power series in ϵ :

$$\psi_f(r) = \psi_i(r) + \epsilon \psi_{s1}(r) + \epsilon^2 \psi_{s2}(r) + O(\epsilon^3). \quad (4)$$

Substituting into

$$\left[-\frac{\hbar^2}{2m} \Delta + \epsilon V \right] \psi_f = E \psi_f \quad (5)$$

and collecting powers of ϵ gives, at order ϵ^0 , the free equation for ψ_i above (automatically satisfied), and at orders ϵ^1, ϵ^2 ,

$$-\frac{\hbar^2}{2m} \Delta \psi_{s1} + V \psi_i = E \psi_{s1}, \quad (6)$$

$$-\frac{\hbar^2}{2m} \Delta \psi_{s2} + V \psi_{s1} = E \psi_{s2}. \quad (7)$$

Writing $K = k$ and rearranging as inhomogeneous Helmholtz equations,

$$(\Delta + K^2) \psi_{s1} = \frac{2m}{\hbar^2} V \psi_i, \quad (8)$$

$$(\Delta + K^2) \psi_{s2} = \frac{2m}{\hbar^2} V \psi_{s1}. \quad (9)$$

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B. Green's Function Solution

Using the outgoing-wave Green's function of the Helmholtz operator, the first-order scattered wave is

$$\psi_{s1}(r) = -\frac{m}{2\pi\hbar^2} \int \frac{\exp(jk|r-r'|)}{|r-r'|} \times V(r') \psi_i(r') d^3r' = \int G_k(r-r') V(r') \psi_i(r') d^3r', \quad (10)$$

with

$$G_k(r) = -\frac{m}{2\pi\hbar^2 r} \exp(jkr), \quad (11)$$

and, iterating,

$$\psi_{s2}(r) = \iint G_k(r-r') V(r') G_k(r'-r'') \times V(r'') \psi_i(r'') d^3r' d^3r''. \quad (12)$$

Formally, $\psi_{sm} = (G_k V)^m \psi_i$ for $m = 1, 2, 3, \dots$, so the exact solution is the Born series

$$\psi_f = \psi_i + \sum_{m=1}^{\infty} \epsilon^m (G_k V)^m \psi_i. \quad (13)$$

For $\psi \in L^2(\mathbb{R}^3)$, boundedness of $G_k V$ follows from

$$\|G_k V \psi\|^2 \leq \left[\int |G_k(r-r') V(r')|^2 d^3r d^3r' \right] \times \left[\int |\psi(r')|^2 d^3r' \right], \quad (14)$$

which is finite provided $\int |V(r')|^2 d^3r' < \infty$ and $\int |G_k(r)|^2 d^3r < \infty$; the latter integral in fact diverges for the free Green's function ($\int r^{-2} \cdot 4\pi r^2 dr = \infty$), but $\|G_k V\|$ can still be made finite for suitable V (e.g. compactly supported or sufficiently rapidly decaying potentials).

C. Resolvent (Operator) Form

Writing $H_0 = -\frac{\hbar^2}{2m} \Delta$, one finds

$$G_k = \frac{2m}{\hbar^2} (\Delta + k^2)^{-1} = -(H_0 - E)^{-1}, \quad (15)$$

so that formally

$$\psi_f = \psi_i + \frac{\epsilon G_k V \psi_i}{I - \epsilon G_k V} = [I - \epsilon G_k V]^{-1} \psi_i. \quad (16)$$

Substituting $G_k = -(H_0 - E)^{-1}$,

$$\begin{aligned} [I - \epsilon G_k V]^{-1} &= [I + \epsilon(H_0 - E)^{-1} V]^{-1} \\ &= [H_0 - E + \epsilon V]^{-1} [H_0 - E] \\ &= [H_1 - E]^{-1} [H_0 - E], \end{aligned} \quad (17)$$

where $H_1 = H_0 + \epsilon V$. This gives

$$[H_1 - E] \psi_f = [H_0 - E] \psi_i. \quad (18)$$

Since $|\psi_i\rangle$ is an eigenstate of H_0 with eigenvalue E , both sides vanish identically and no information is obtained this way — the resolvent $[H_0 - E]^{-1}$ is singular exactly at the energy of the incident wave, so this naive manipulation is degenerate.

D. Resummed Born Series

Instead, resum the series directly. Using

$$\begin{aligned} G_k \sum_{m=1}^{\infty} \epsilon^m (V G_k)^{m-1} V &= \epsilon G_k [I - \epsilon V G_k]^{-1} V \\ &= \sum_{m=1}^{\infty} \epsilon^m (G_k V)^m, \end{aligned} \quad (19)$$

and

$$I - \epsilon V G_k = I + \epsilon V (H_0 - E)^{-1} = [H_1 - E] [H_0 - E]^{-1}, \quad (20)$$

one obtains

$$\sum_{m=1}^{\infty} \epsilon^m (G_k V)^m = \epsilon G_k [H_0 - E] [H_1 - E]^{-1} V. \quad (21)$$

Using $G_k [H_0 - E] = -(H_0 - E)^{-1} (H_0 - E) = -I$, this simplifies to

$$\sum_{m=1}^{\infty} \epsilon^m (G_k V)^m |\psi_i\rangle = -\epsilon [H_1 - E]^{-1} V |\psi_i\rangle, \quad (22)$$

so that

$$|\psi_f\rangle = [I - \epsilon [H_1 - E]^{-1} V] |\psi_i\rangle. \quad (23)$$

Check: multiplying (23) by $H_1 - E$ gives

$$(H_1 - E) |\psi_f\rangle = (H_1 - E - \epsilon V) |\psi_i\rangle = (H_0 - E) |\psi_i\rangle, \quad (24)$$

and both sides vanish, consistently.

E. Perturbative Expansion

To expand (23) order by order in ϵ , write

$$(H_1 - E)^{-1} = (H_0 - E + \epsilon V)^{-1} \quad (25)$$

$$\begin{aligned} (H_1 - E)^{-1} &= [I + \epsilon(H_0 - E)^{-1} V]^{-1} \\ &\times (H_0 - E)^{-1} \\ &= [I - \epsilon(H_0 - E)^{-1} V] (H_0 - E)^{-1} \\ &+ O(\epsilon^2). \end{aligned} \quad (26)$$

Hence

$$\begin{aligned} |\psi_f\rangle &= |\psi_i\rangle - \epsilon [H_0 - E]^{-1} V |\psi_i\rangle \\ &+ \epsilon^2 [H_0 - E]^{-1} V [H_0 - E]^{-1} V |\psi_i\rangle + O(\epsilon^3) \\ &= [I + \epsilon G_k V + \epsilon^2 (G_k V)^2] |\psi_i\rangle + O(\epsilon^3), \end{aligned} \quad (27)$$

recovering the Born series of Sec. II-B, as it must.

F. Møller Wave Operators

Let $|\psi_0\rangle$ denote the free (H_0 -evolved) output state and $|\psi_f\rangle$ the state evolving under H_1 . As $t \rightarrow +\infty$ the two must coincide up to phase:

$$\lim_{t \rightarrow \infty} \|e^{-itH_1} |\psi_f\rangle - e^{-itH_0} |\psi_0\rangle\| = 0, \quad (28)$$

so $|\psi_f\rangle = \Omega(t) |\psi_0\rangle$ with $\Omega(t) = \lim_{t \rightarrow \infty} e^{itH_1} e^{-itH_0}$. Since

$$\begin{aligned} \frac{d}{dt} (e^{itH_1} e^{-itH_0}) &= i e^{itH_1} (H_1 - H_0) e^{-itH_0} \\ &= i \epsilon e^{itH_1} V e^{-itH_0}, \end{aligned} \quad (29)$$

integrating from 0 to t and letting $t \rightarrow \infty$,

$$\Omega_+ = I + i\epsilon \int_0^\infty e^{itH_1} V e^{-itH_0} dt. \quad (30)$$

Using the spectral resolution $H_0 = \int_{\mathbb{R}} \lambda dE(\lambda)$ and

$$\int_0^\infty e^{it(H_1 - \lambda + i\delta)} dt = i(H_1 - \lambda + i\delta)^{-1} \quad (31)$$

(convergent for $\delta > 0$, then $\delta \rightarrow 0^+$),

$$\Omega_+ = I - \int_{\mathbb{R}} (H_1 - \lambda + i\delta)^{-1} V dE(\lambda), \quad (32)$$

and for an input state of definite energy E , $dE(\lambda)|\psi_0\rangle = \delta_{E,\lambda}|\psi_0\rangle$, giving

$$|\psi_f\rangle = |\psi_0\rangle - \epsilon[H_1 - E + i\delta]^{-1} V |\psi_0\rangle. \quad (33)$$

By the analogous argument as $t \rightarrow -\infty$,

$$|\psi_f\rangle = \Omega_- |\psi_i\rangle, \quad \Omega_- = I - \epsilon[H_1 - E - i\delta]^{-1} V, \quad (34)$$

so that, writing $\Omega_\pm$ as functions of energy,

$$\Omega_+(E) = I - \epsilon[H_1 - E + i\delta]^{-1} V, \quad (35)$$

$$\Omega_-(E) = I - \epsilon[H_1 - E - i\delta]^{-1} V. \quad (36)$$

The Born approximation of Sec. II-B is the finite-order truncation of $|\psi_f\rangle = \Omega_-(E)|\psi_i\rangle$.

G. Scattering Matrix

Since $\Omega_+(E)$ is an isometry, $\Omega_+(E)^* \Omega_+(E) = I$, so $|\psi_0\rangle = \Omega_+(E)^* |\psi_f\rangle = \Omega_+(E)^* \Omega_-(E) |\psi_i\rangle$, which defines the scattering matrix at energy E ,

$$S(E) = \Omega_+(E)^* \Omega_-(E), \quad |\psi_0\rangle = S(E) |\psi_i\rangle. \quad (37)$$

$S(E)$ is unitary on the space of fixed-energy states, i.e. states defined on the unit sphere, $L^2(S^2)$. To leading order in ϵ ,

$$\begin{aligned} S(E) &= [I - \epsilon[H_1 - E + i\delta]^{-1} V]^* \\ &\quad \times [I - \epsilon[H_1 - E - i\delta]^{-1} V] \\ &= I - \epsilon \left\{ V[H_0 - E - i\delta]^{-1} \right. \\ &\quad \left. + [H_0 - E - i\delta]^{-1} V \right\} + O(\epsilon^2). \end{aligned} \quad (38)$$

H. Plane-Wave Matrix Elements

Take two incident plane waves

$$\psi_{i1}(r) = C_1 \exp(jk\hat{n}_1 \cdot r), \quad (39)$$

$$\psi_{i2}(r) = C_2 \exp(jk\hat{n}_2 \cdot r). \quad (40)$$

Their Fourier transforms are

$$\hat{\psi}_{i1}(k') = (2\pi)^{3/2} C_1 \delta(k' - k\hat{n}_1), \quad (41)$$

$$\hat{\psi}_{i2}(k') = (2\pi)^{3/2} C_2 \delta(k' - k\hat{n}_2), \quad (42)$$

so that

$$\begin{aligned} \langle \psi_{i2} | \psi_{i1} \rangle &= \int \bar{\psi}_{i2}(k') \psi_{i1}(k') d^3 k' \\ &= (2\pi)^3 \bar{C}_2 C_1 \delta(k(\hat{n}_2 - \hat{n}_1)) \\ &= \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1), \end{aligned} \quad (43)$$

and

$$\begin{aligned} \langle \psi_{i2} | V | \psi_{i1} \rangle &= \bar{C}_2 C_1 \int V(r) \\ &\quad \times \exp(-jk(\hat{n}_2 - \hat{n}_1) \cdot r) d^3 r \\ &= (2\pi)^{3/2} \bar{C}_2 C_1 \hat{V}(k(\hat{n}_2 - \hat{n}_1)). \end{aligned} \quad (44)$$

By the Riemann–Lebesgue lemma, $\lim_{k \rightarrow \infty} \hat{V}(k(\hat{n}_2 - \hat{n}_1)) = 0$, so this term is negligible at high energy. The Born term is

$$\begin{aligned} \langle \psi_{i2} | [H_0 - E]^{-1} | \psi_{i1} \rangle &= -\frac{2m}{\hbar^2} \langle \psi_{i2} | [\Delta + k^2]^{-1} V | \psi_{i1} \rangle \\ &= \frac{m}{2\pi\hbar^2} \int \frac{\exp[jk|r - r'|]}{|r - r'|} \\ &\quad \times \bar{\psi}_{i2}(r) V(r') \psi_{i1}(r') d^3 r d^3 r'. \end{aligned} \quad (45)$$

Since ψ_{i1}, ψ_{i2} are concentrated at $r \rightarrow \infty$ while $V(r)$ is concentrated near the origin, $\langle \psi_{i2} | V | \psi_{i1} \rangle \approx 0$, and

$$\begin{aligned} \langle \psi_{i2} | S(E) | \psi_{i1} \rangle &\approx \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1) \\ &\quad - \frac{m\epsilon}{2\pi\hbar^2} \bar{C}_2 C_1 \int \frac{\exp(jk|r - r'|)}{|r - r'|} \\ &\quad \times \exp(jk[\hat{n}_1 \cdot r' - \hat{n}_2 \cdot r]) V(r') d^3 r' d^3 r. \end{aligned} \quad (46)$$

In the far field, $r \gg r'$, so

$$\frac{\exp(jk|r - r'|)}{|r - r'|} \approx \frac{e^{jkr}}{r} \exp(-jk\hat{r} \cdot r'), \quad (47)$$

giving

$$\begin{aligned} \langle \psi_{i2} | S(E) | \psi_{i1} \rangle &\approx \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1) \\ &\quad - m\epsilon \bar{C}_2 C_1 (2\pi)^{3/2} \int \hat{V}(k|\hat{r} - \hat{n}_1|) \\ &\quad \times \frac{\exp[jk(r - \hat{n}_2 \cdot r)]}{r} d^3 r. \end{aligned} \quad (48)$$

Without the far-field approximation, the corresponding (unapproximated) expression is

$$\begin{aligned} \langle \psi_{i2} | S(k) | \psi_{i1} \rangle &\approx \frac{(2\pi)^{3/2} \bar{C}_2 C_1}{k} \delta(\hat{n}_2 - \hat{n}_1) \\ &\quad - \frac{m\epsilon}{2\pi\hbar^2} \bar{C}_2 C_1 \int \frac{\exp(jk|r - r'|)}{|r - r'|} \\ &\quad \times \exp(jk[\hat{n}_1 \cdot r' - \hat{n}_2 \cdot r]) V(r') d^3 r' d^3 r. \end{aligned} \quad (49)$$

III. CONCLUSIONS

We have derived, from first principles, the full Born-series expansion of the scattered state ψ_f , its resummation into the exact resolvent form $|\psi_f\rangle = [I - \epsilon(H_1 - E)^{-1} V] |\psi_i\rangle$, the Møller wave operators $\Omega_\pm(E)$, and the resulting energy-shell scattering matrix $S(E) = \Omega_+(E)^* \Omega_-(E)$, which is unitary on $L^2(S^2)$. We also obtained the plane-wave matrix elements $\langle \psi_{i2} | S(E) | \psi_{i1} \rangle$ to first order in the coupling ϵ , both with and without the far-field approximation.

The unitarity of $S(E)$ on $L^2(S^2)$ is the key fact that motivates treating it as a candidate quantum gate: if a discrete set of incident directions $\{\hat{n}_1, \dots, \hat{n}_N\} \subset S^2$ is chosen to represent an N -dimensional computational basis, the restriction of $S(E)$ to the span of $\{|\psi_{i,\ell}\rangle\}_{\ell=1}^N$ is, to the order computed here, the matrix with entries

$$U_{\ell\ell'} = \langle \psi_{i\ell} | S(E) | \psi_{i\ell'} \rangle, \quad (50)$$

built from the expressions of Sec. II-G. Realising a specific target gate U (e.g. a Hadamard or CNOT-type matrix in an appropriate basis) then becomes an inverse problem: choose $V(r)$, the incident directions $\hat{n}_\ell$, and the energy E so that $U_{\ell\ell'}$ matches the target gate to within the desired precision.

Three points are left open by the present derivation and are the natural next steps. First, the matrix $U_{\ell\ell'}$ above is exactly unitary only when the full (non-perturbative) $S(E)$ is used; the first-order expression derived here is unitary only up to $O(\epsilon^2)$, so either higher orders must be kept or V must be restricted to a regime where the first-order truncation is an adequate approximation to a genuinely unitary evolution. Second, no explicit potential $V(r)$ or explicit target gate has yet been worked through numerically; doing so — for instance for a spherically symmetric $V(r)$ and a two-direction encoding representing a single qubit — would be the natural next section of this work. Third, the delta-function term $\delta(\hat{n}_2 - \hat{n}_1)$ in $\langle \psi_{i2} | S(E) | \psi_{i1} \rangle$ is only well defined in a discretised or regularised sense once a finite set of directions is chosen, and this regularisation should be made explicit before any numerical gate fidelity is computed.

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